Group Name: 04

Team

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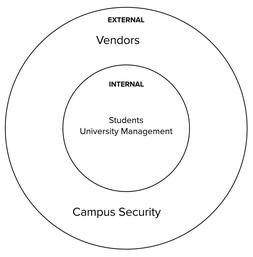
Stakeholder Map



Research Plan

Research Question: Overcrowding at the university's single turnstile entrance causing significant delays for students

Research Participants: Who are the key stakeholders you need to engage with to learn more about your research question? Narrow down your stakeholders and make a list of minimum 6 people you will engage with (keeping in mind if they are easy to access). Think about 'extreme users' that might give you wide insights into the problem. This list should be of specific types of people you will be approaching e.g. 1) SDP Freshman Student, 2) OAP Advisor.

1)
2)
3)
4)
5)
6)

Research Methods: Decide which primary research methods would be best to use to get rich insights into your problem area, given the time you have. Will you use different methods for different stakeholders? Can you gather any secondary research on this matter - what data, reports, case studies, surveys might exist already within the University or elsewhere on this topic? Write down your chosen research methods here.

Research Materials: Write down any research materials you need to create in order to carry out your research methods e.g. interview guides for interviews/contextual inquiries, cultural probes, diary studies. You can use the space on the right to create the materials.

Personas

Create your persona(s). You need to create at least one persona.

Name Ahsan Shah

About
Short bio about them e.g. age, gender, profession, major, socio-economic background

Age: 19
Gender: Male
Major: Electrical Engineering

Studies at Habib University and has to travel for 2 hours daily, via personal conveyance, to reach the university by 8:15 for a class. Doesn't like huge crowds.

Goals
What does your persona hope to accomplish?

- Arrive on time for lectures and maintain at least 85% attendance.

Quote
A memorable part of your research that encompasses the essence of their experience, needs, wants.

"Challenges are stepping stones to success"

Needs
What do they need in order to successfully accomplish their goals?

Faster entry process
Shorter or no queues
Better functioning security

Frustrations/challenges
What are the challenges this persona faces? What barriers are they facing in getting towards their goal?

- Long queues in the morning
- Gets late to class due to the queue at gate 4
- Medical conditions restrict him to stand for longer periods of time.

Name Ahmed Bilal

About
Short bio about them e.g. age, gender, profession, major, socio-economic background

Age: 55
Gender: Male
Profession: Security Incharge

Works as a senior security manager at Habib University.

Goals
What does your persona hope to accomplish?

-Security of students and faculty.
-General security of campus.
-Easy and fast entrance of students in the campus without compromising on their security.

Quote
A memorable part of your research that encompasses the essence of their experience, needs, wants.

"We will never compromise on our students or staff security"

Needs
What do they need in order to successfully accomplish their goals?

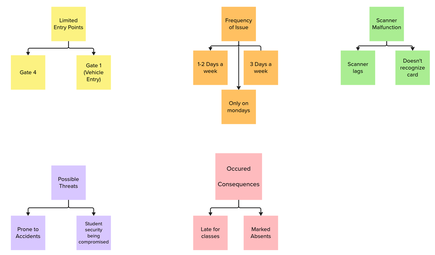
- They need to manage the entry and exit points which will not compromise on student and campus security. Traffic management is a very hard thing on campus, and they are currently working on it.
- They need students cooperation in order todo sort things out in a proper manner

Frustrations/challenges
What are the challenges this persona faces? What barriers are they facing in getting towards their goal?

-Due to mutual class timings, gate 4 gets crowded due the excessive number of students wanting to enter in a short time span.
-Students do not abide by the rules of accessing the institute, for example, students forgetting their ID cards and then either disturbing the flow of entrance or getting in to an argument with the security personnel.
-Limited entry points

Affinity Mapping

Create clusters / Themes from your Raw Data on the white boards and share the summary here. You can bring all your relevant sticky notes from the white board on this paper. Make sure each cluster/theme is labeled with a heading that describes it.



Insight Statements

Write out your key insights here. You should turn each cluster/theme into an insight statement. Use the insight formulas to help frame clear, coherent and actionable insights.

1. Students get late to their classes because of longer queues at gate 4.
2. Security wants to ensure safety of the students but the long queues pose difficulties
3. Students want to access the university quicker by the getting an alternative entrance to the university.

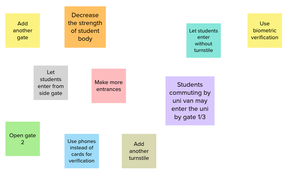
selected statement : 3

HMW and Ideation

1 Re-write your chosen How Might We (HMW) question here.

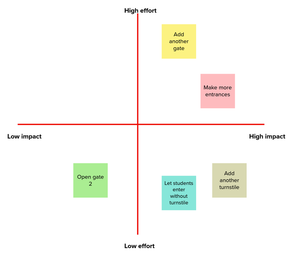
Q, How might we make the entrance process smoother?
Q, How might we reduce the queues early morning?
Q, How might we eradicate the queue system at the gate?
Q, How might we make the entrance process quicker?
Q, How might we ensure the student security while maintaining process efficiency?
Q, How might we work around the already existing entrances to resolve the solution without much hassle or investment?

2 This is your idea board. Capture all of the ideas you have generated from your individual brainstorm, change cards, silent circuit, lateral inspiration and local/global examples. Make sure the writing on your post-it notes are legible.



3 Put the idea with the highest votes here

4 Create your Impact vs Effort matrix. Plot the ideas that received the most dot votes against this matrix.



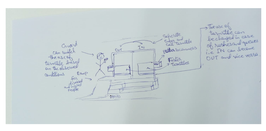
5 Write your one chosen idea/concept here, describing it in 2-3 sentences. Ideally this should be a high impact/low effort idea. You will then take this idea forward and start prototyping it in the next class.

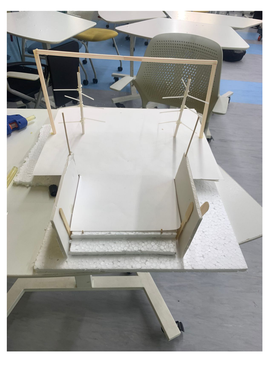
-->Add another turnstile.
-->Enter without using turnstile.

Either the university adjusts te entrance to accomodate two turnstiles so that the queue divides into two, or, turnstiles are removed as an whole and students can just enter by verification.

Low-Fidelity Prototyping

1 Create / Attach your initial idea/concept sketches.





2 Re-visit your persona(s). What would need to be included in your idea to work for them and meet their needs? **Write out the key components or features of your idea with this in mind.**

The main issue at gate 4 is that there is only one turnstile which makes the entrance process slower. if we add another turnstile at gate 4, more people will be able to enter at the same time and there will be no queues or traffic at the gate.

3 Write a Short summary about the nature and functionality of your low fidelity prototype

We are planning to add another turnstile, but due to less space we planned to move turnstile a little forward where there is possible to add another turnstile. after talking to the security we can say that it is possible to do so.
in this way students wont be waiting in long queues, and more people will be able o enter university without compromising security.

Testing Round 1

Document your work from 21st-Oct-2024

1 Riskiest Assumptions

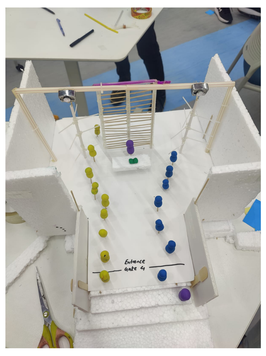
Assumptions are notions that need to be proved to be true for your idea to work. If they are challenged or disproved, you will need to change aspects of your idea/ prototype. **What are the riskiest assumptions about your idea/prototype?**

1. No one is exiting the university when there is a lot of traffic entering the university. or vice versa

2. there are no such situation happening which might compromise security of students standing outside.

2 Who did you test with? Write down a list of people/users/stakeholders you spoke to.

1. students who enter from gate 4
2. head of security.
3. security guard at gate 4.



3 Feedback & Learnings From Testing

What worked?

According to the head of security, it is a very good and practical idea which could be implemented. It does not compromise on the security of campus and students.

As per student's response, this will actually help them in entering faster so that they wont be getting late for classes, especially in rush hours.

What didn't?

We need to plan places of each thing for entrance, for example: security guard seat, kitchen door, normal entrance door(without card).

Were any of your assumptions confirmed or disproved? How do you know?

One of our assumptions was approved, that nobody is usually exiting the university at rush hours. usually in morning students are entering only, and in evening they are usually exiting.

Our assumption of no unforeseen conditions was disapproved by the head of security as they said there is serious threat to the campus mainly because of foreign faculty and other VIP personality.

What new questions or ideas arose?

Make a better and even more detailed design.
Bio-Metric entrance instead of cards so entrance process gets a bit more faster.

What do you need to change going forward?

1. According to the head of security, normal gate(without card) should be removed from gate 4. people who are on wheelchairs will be entering from somewhere else.

Higher-Fidelity Prototype

Describe your high fidelity prototype and attach any relevant details here.

Testing Round 2

Document your work from : Date __/__/____

1 Who did you test with? Write down a list of people/users/stakeholders you spoke to.

2 Feedback & Learnings From Testing

What worked?

What didn't?

Were any of your assumptions confirmed or disproved? How do you know?

What elements meet the needs of your persona? What don't?

What new questions or ideas arose?

3 What do you need to change going forward?

Summarize the key changes you would make to the design of your solution, if you were to carry on with the testing and learning process.
If you were to create the next iteration of your prototype based on the feedback learnt, what changes would you want to implement? Give as much detail here as possible.